TrustLens AI

AI-Powered Product Condition Verification
for Safer E-Commerce

Team:
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Mentor:
Mr. Shivanshu Upadhyay

Problem & Solution Overview

Problem

- Vague damage labels
- Buyers can't judge usability
- Unfair pricing
- Fake/manipulated images
- High dispute rates
- Manual verification is inefficient

Our Solution

✓ **End-to-end AI-powered Damage Detection**
✓ **Clear explanations (NLP)**
✓ **Fair price recommendations**
✓ **Automated verification**

Objectives & Innovation

Project Objectives

1. Build buyer trust in damaged products
2. Provide clear AI explanations
3. Enable fair pricing decisions
4. Detect & prevent image fraud
5. Reduce disputes & returns
6. Create scalable trust layer

Unique Features & Innovation

🎯 **Multi-AI Integration**
YOLOv8 + NLP

💡 **Clear Explanations**
Human-readable AI reports

🔍 **Visual Proof**
Severity-to-discount mapping
Damage visualization on images

💰 **Rule-Based Pricing**

🚀 **Fast & Scalable**
<5s processing, cloud-based

Key Features & Capabilities

AI Damage Detection

YOLOv8-powered visual analysis detecting scratches, cracks, dents with 80%+ accuracy

≥80% Accuracy

Rule-Based Pricing

Predefined severity-to-discount rules with transparent tier-based pricing logic

±10% Accuracy

Clear Explanations

NLP-powered human-readable reports explaining damage impact on functionality

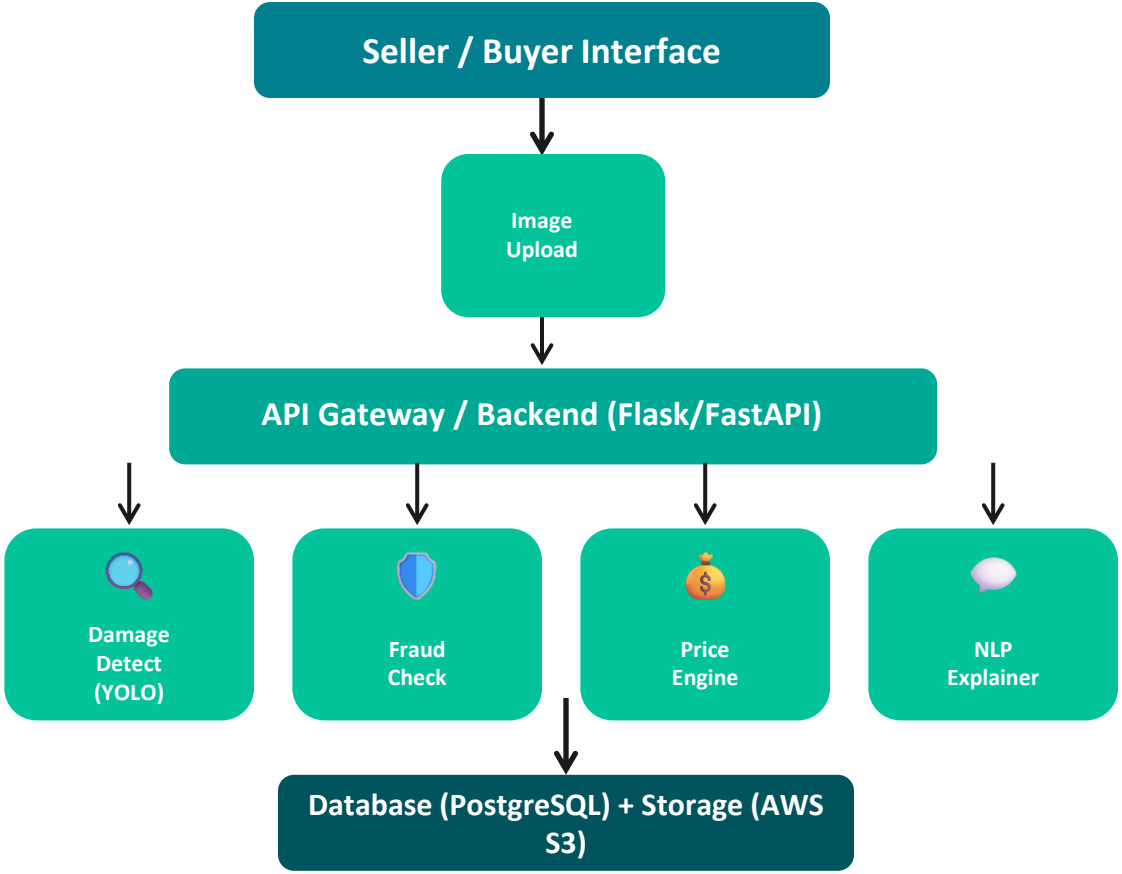
≥85% Clarity

Fast Processing

Complete analysis pipeline from upload to report in under 10 seconds

<10s Total

System Architecture & Workflow



Technical Approach & Implementation

1. Damage Detection (YOLOv8)

- Pre-trained YOLOv8 model fine-tuned on damaged product dataset
- Detects damaged regions with bounding boxes
- Classifies damage type (scratch, crack, dent, etc.)
- Severity scoring (1-10 scale) based on size and location

2. Fraud Detection (Multi-Layer)

- Perceptual hashing (pHash) for duplicate detection
- EXIF metadata analysis for editing traces
- Error Level Analysis for digital manipulation
- Database comparison against known fraud images

3. NLP Explanation (LLM)

- Claude API / GPT-4 for natural language generation
- Converts technical damage data to buyer-friendly text
- Interactive chatbot for product questions
- Severity-to-impact language mapping

4. Rule-Based Pricing & Trust

- Severity-to-discount rule mapping (e.g., 5% per severity point)
- Predefined discount tiers based on damage categories
- Seller trust score (listing accuracy + dispute rate)
- Transparent pricing with clear justification

Uniqueness & Competitive Advantage

Feature	Competitors	Our Platform
AI Damage Detection	❌	✅ 80%+ Accuracy
Visual Damage Proof	❌	✅ Bounding Boxes
Fraud Prevention	⚠️ Limited	✅ Multi-layer (75%+)
Clear Explanations	❌	✅ NLP-powered
Rule-Based Pricing	⚠️ Manual	✅ Automated
Trust Scoring	❌	✅ Seller Metrics
Processing Speed	⚠️ Slow	✅ <10s

Team Roles & Contributions

1

Kunal Singhal

Team Leader | Backend & AI/ML

Key Responsibilities:

- Project planning & coordination
- AI model development (YOLOv8)
- Backend API development (FlastApi)

Technical Skills:

Python, PyTorch, FastApi, PostgreSQL
Additional: Database design (PostgreSQL), System integration

2

Kartik Agrawal

Frontend Developer | UI/UX & Deployment

Key Responsibilities:

- React.js frontend development
- UI/UX design (Tailwind CSS)
- Cloud deployment (AWS/Heroku)

Technical Skills:

React.js, Tailwind, Docker, AWS
Additional: CI/CD pipeline setup, Performance optimization

3

Manish Bhardwaj

AI/ML Developer | ML & Testing

Key Responsibilities:

- Model training & fine-tuning
- Fraud detection implementation
- NLP integration

Technical Skills:

PyTorch, Computer Vision, NLP
Additional: Testing & validation, Model optimization

Expected Impact

Business Impact

- Increase buyer confidence in refurbished products
- Fair & transparent pricing for damaged items
- Reduce fraudulent listings by 75%+
- Decrease disputes & returns by 50%+
- Automated verification (100x faster than manual)
- Scalable solution for any e-commerce platform

Expected Outcomes

✓ Reduced e-commerce trust gap through automation
✓ Scalable, cloud-deployed solution ready for real-world use
✓ Comprehensive documentation & testing (≥70% code coverage)

Implementation Roadmap & Milestones

Phase 1-2

Complete

Planning & Setup

- Requirements gathering
- Architecture design
- Dataset collection
- Tech stack setup

Phase 3-6

In Progress

Core Development

- Damage detection model
- Fraud detection
- NLP integration
- Backend APIs

Phase 7-9

Planned

Integration

- Frontend development
- System integration
- Trust scoring
- Price engine

Phase 10-13

Planned

Testing & Launch

- Full testing
- Bug fixes
- Deployment
- Documentation

Challenges & Future Enhancements

Key Challenges

- Dataset quality & diversity
- Model accuracy for complex damage
- LLM API cost optimization
- Real-time processing requirements
- Integration complexity

Future Enhancements

- Multi-language support (Hindi, regional)
- Mobile app (iOS & Android)
- Video-based damage analysis
- Integration with major platforms
- Advanced analytics dashboard

Risk Mitigation Strategies

• **Insufficient data:**	Data augmentation, transfer learning, multiple sources
• **Model accuracy:**	Ensemble methods, extensive validation, fine-tuning
• **API costs:**	Caching, prompt optimization, usage monitoring
• **Timeline delays:**	Weekly milestones, scope prioritization, buffer time

Project Summary

TrustLens AI

Building Trust in E-Commerce Through AI Innovation

✓ AI-Powered Damage Detection (≥80% accuracy)
✓ Human-Readable Explanations (≥85% clarity)
✓ Seller Trust Scoring & Analytics
✓ Fast Processing (<10s) & Cloud Deployment

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